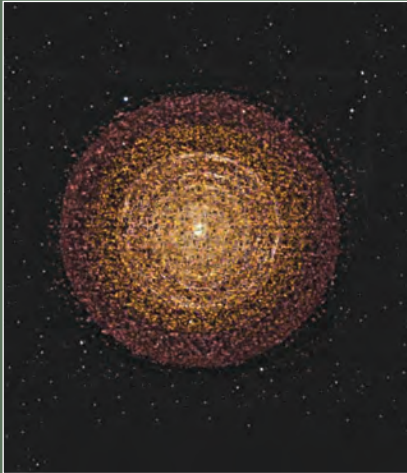
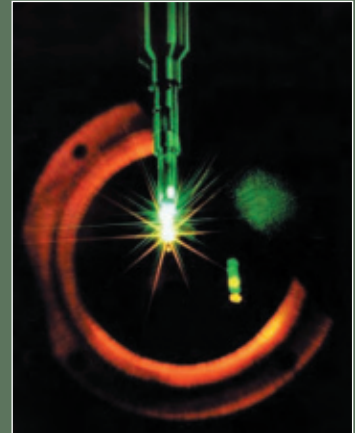


Technology that will save the world

Are your nerves on end in anticipation of an oncoming apocalypse? Do you have that nagging feeling that civilisation has gone too far? Environmental alarmists, religious historians and doomsday prophets are all predicting the end of the world. Fear not! Technology is here to rescue the planet. Most of these technologies are well beyond the scope of current science, but the dreams of these technologies, and what they promise may just about distract us from the constant twenty first century nuclear backdrop.

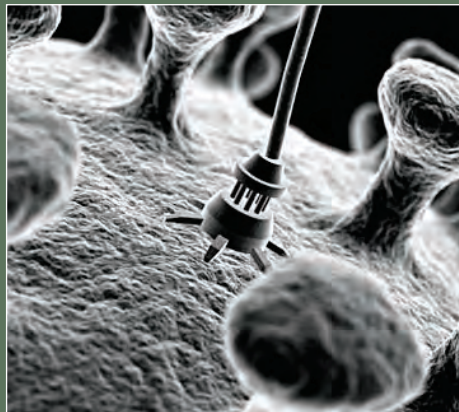


MATRIOSHKA BRAIN: The Matrioshka Brain is a device that can swallow up and use the entire output of a star. Although this can be achieved using a large array of satellites, the most common model is a shell of energy absorbing material all around the star. This is very similar to a Dyson Sphere, which is basically a star enclosed within a solar panel - but the Matrioshka Brain goes one step farther. The Brain uses the entire energy output of a star as the energy source for a computer. That's right, the Matrioshka Brain is a star sized computer. If mankind ever brings itself to make one of these, we are pretty sure that the designers will throw in too few USB ports, and even this computational beast would need periodic RAM upgrades. We're kidding, actually, scientists are hard pressed to find any uses for so much computation power. Start up intergalactic Twitter though, and the fail whale may still appear.



NUCLEAR FUSION: This one has been a tough nut to crack. If and when we manage to contain and sustain Nuclear Fusion for appreciably long amounts of time, we will be opening a pandora's box of sorts. While not exactly giving us infinite energy, it would put in a lot of juice into our power grids. China will most definitely be able to build space lasers, Japan can go about powering giant mechanical flying robotic exoskeletons, and all the CRT monitors still lying about offices in India can continue on their long, and slow death. The best bet we have so far is the Joint European Torus (JET), which spat out 10MW of fusion power over 0.5 seconds way back in 1997. Or, we could just get the Chinese to start on the dyson sphere already. Pictured above is a "microsun" experiment.

NANOTECHNOLOGY: The scientists fooling around with small things are indulging in spine chilling experiments already. Fresh from the labs report mosquitoes that inject vaccines instead of diseases, and tiny robotic doctors that administer gene therapy from inside individual cells. Using light, atomic or graphene based computing would be a quantum leap forward in terms of computational power, but the real scope of the emerging technologies are in the spheres of materials, and biotechnology. Nanotechnology is the kind of disruptive progress that has the potential to drastically affect all other fields.



THE INTERNET: If you think the internet has come far, think again. The true potential of the internet is not even being hinted at, at this point of time. They are trying to put an IP address into everything, including, say the box in which a router comes in. Right now, only humans are using the internet. Pretty soon, refrigerators, hospitals and the bowl used to feed milk to the stray cat could all tweet a shopping list to the local grocery store. We're thinking "in-your-face" in a more literal sense as well, maybe your skull will have an IP address too, and augmented reality overlays beamed to your visual cortex. The Internet after a 1000 years would be wild and exotic.

FASTER THAN LIGHT TRAVEL: Don't worry, Einstein won't really have a problem with this, despite of what you might have heard. Einstein convincingly argued that nothing can be accelerated faster than light. That does not automatically rule out faster than light travel though. There are supposed particles that travel faster than light to begin with. Also, some scientists working with obscure branches of mathematics and physics claim that if you come up with a device to bend or warp the space-time around the craft, then it can travel large distances without accelerating faster than light. The loophole is that you bring Z to where A is, instead of taking the effort of hauling yourself from A to Z. Neat huh? (PS: Wormholes won't work, matter does not survive singularity the way it went in).

